Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2403)

Key

SECTION – I

1.	A	7.	B	13.	D	19.	3412
2.	A	8.	A	14.	A	20.	3124
3.	D	9.	B	15.	D	21.	A
4.	C	10.	C	16.	D	22.	1234
5.	D	11.	B	17.	B	23.	D
6.	B	12.	B	18.	B	24.	C

SECTION – II

1.	6	6.	58	11.	2	16.	A
2.	C	7.	C	12.	A	17.	D
3.	17	8.	53	13.	11	18.	C
4.	B	9.	D	14.	D	19.	D
5.	C	10.	B	15.	C	20.	A

SECTION – III

1.	5	7.	252	13.	A	19.	B
2.	D	8.	100	14.	B	20.	0.25
3.	D	9.	C	15.	D	21.	A
4.	21	10.	C	16.	A	22.	4321
5.	0	11.	88	17.	4		
6.	C	12.	C	18.	C		

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 544

1. Option A: This cannot be concluded from 'If the precise function of ritual jades in the late Neolithic is indeterminate, such is not the case in the Bronze Age.' We have evidence of the use of ritual jade in the Neolithic era. We just don't know the purpose. Hence, Option A is the answer.

Option B: From the information that Shang craftsmen were excellent at understanding the jade working tradition and the technicalities involved, and that during the Bronze Age (overlapping with the Shang dynasty) jade usage was well-documented, we can conclude that there is written record/evidence of the use of jade in the Shang era. Hence, Option B is not the answer.

Option C: This can be concluded from the written records which showed that '***jades were used in sacrificial offerings to gods and ancestors, in burial rites, for recording treaties between states, and in formal ceremonies at the courts of kings***.' Most of these can be considered official uses. Hence, Option C is not the answer.

Option D: This can be understood from the data that Shang craftsmen understood the technical language of the jade working tradition '***developed in the diverse late Neolithic cultures.'*** Also, while jade usage in Bronze Age was well-documented, the same could not be said for the Neolithic age. So, adding these two together, we could say that the Neolithic era preceded the Bronze Age. Option D is not the answer.

Choice (A)

2. Option A: As understood from the passage the taotie and other decorative motifs were used widely but their purpose is unknown. This is confirmed by 'Currently, the significance of the taotie, as well as the other decorative motifs, in Shang society is unknown.' So, the purpose of their usage could be a good area of investigation. Hence, Option A is the answer.

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Option B: The end of the Western Zhou witnessed the introduction of iron in China. Also, the Shang and Zhou dynasty eras were called the Bronze Age because of the prevalence of bronze. So, it can be inferred that Bronze was not the most preferred material after the Zhou dynasty era was past. Option B is not the answer.
Option C: Since the passage doesn't touch upon the cost of the materials or how they were valued, whether it was jade or bronze, this question, in the context of the passage doesn't serve much academic purpose. Hence, Option C is not the answer.
Option D: This has already been answered in the passage. The passage mentions that 'Shang craftsmen had full command of the artistic and technical language developed in the diverse late Neolithic cultures that had a jade-working tradition.' So, it can be understood that late Neolithic cultures had a jade-working tradition instead of bronze metalworking. Option D is not the answer. Choice (A)

3. Option A: It can be understood that iron appeared only during the Eastern Zhou dynasty. This dynasty was prevalent between 770 – 256 BC. So, Option A is true.
Option B: Bronze played an important role in the material culture of the time, and it was used for ritual vessels as well as for weapons and chariots. That is why the era of Shang and Zhou dynasties was called the Bronze Age according to the third para of the passage. Option B is not the answer.
Option C: The piece mold casting was less popular as understood from the observation that the lost-wax method was used by all other Bronze Age cultures. However, this process was used so that the metalworkers could achieve better definitions of designs and detail. Option C is not the answer.
Option D: The passage says that 'Shang craftsmen had full command of the artistic and technical language developed in the diverse late Neolithic cultures that had a jade-working tradition.' However, it has not been mentioned that bronze craftsmen were inspired by jade-working techniques. Hence, Option D is the answer. Choice (D)

Option D: This is not true, since Spring and Autumn period, and Warring States period are not separate from the Zhou period. They are part of the Eastern Zhou dynastic rule. Hence, Option D is not the answer. Choice (C)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 517

5. The Pandora's box refers to a bunch of unexpected problems spawned by a trigger, in this case the finding that notes that cloud loss might result in temperature volatility.

Option A: This is one of the potential disasters that scientists feel could result from increasing temperatures. Increasing temperatures cause cloud loss which could cause temperatures to spike up further. Option A is not the answer.
Option B: Cloud loss causes unexpected issues because clouds help blanket the earth partially from heat. Absence of clouds causes faster heating turning the increase in temperatures into a spiralling situation that can be out of control. This is one of the potential climate surprises caused by excessive warming as understood from the finding. Option B is not the answer.
Option C: Warming temperatures would lead to a loss of clouds which would lead to an additional increase in temperature again because of the role clouds usually play in reducing heating of the environment. So, this is one of the possible scenarios that lead to unexpected climate changes. Option C is not the answer.
Option D: High CO2 emissions lead to increase in environmental temperatures. This is expected and it is not one of the pandora's box situations. It is already known and accounted for. The unexpected climate change is because of the tipping point the author talks about where the CO2 levels cause a point of no return spawning a new set of issues, possibly because of cloud loss. Hence, Option D is the answer. Choice (D)

4. Option A: Spring and Autum are not two separate periods. Also, Warring States and Qin are two different periods, with the Qin unifying China and taking the power at the end of the Warring States period. Option A is not the answer.
Option B: The Warring States period was part of the Eastern Zhou period. It was the second sub-division of the Eastern Zhou dynastic rule at the end of which the Qin dynasty took over China. Option B is not the answer.
Option C: This is the correct sequence of periods covered in the passage. First, it was the Shang dynasty, followed by Western Zhou. The Eastern Zhou period is divided into parts: Spring & Autumn period and Warring States period. At the end of the latter, the Qin dynasty's rule had started. Option C is the answer.

6. Option A: The statement made by Kerry Emanuel is that 'scientists must now make an effort to independently replicate the work.' That means others must do similar research and check if the findings can be replicated. It is not a question of whether these results have been reviewed (in fact, the publishing in a journal indicates they have been reviewed). Option A is not the answer.
Option B: The need for scientists to independently replicate the findings comes from the idea that it should be provable for the conclusions to stick. So, cloud loss due to warming should be proven by multiple simulations and that too by other scientists and that is what has led Kerry Emanuel to make these suggestions. Option B is the answer.

Option C: The findings are about how cloud loss caused by earth's warming will have a further cascading effect on the temperatures. So, Kerry Emanuel's point is not on the basis that we need to establish, firstly, a corelation between cloud loss and warming of the planet. Option C is not the answer.
Option D: This is the statement made by Huber and it has nothing to do with the suggestion made by Kerry Emanuel. Huber already agrees that the tipping point may explain temperature volatility, but his agreement is irrelevant to Emanuel's suggestion that the study must be repeated multiple times to be sure. Hence, Option D is not the answer. Choice (B)

7. Option A: The PETM wasn't mentioned just to discuss the cataclysmic event from 56 million years ago. The author mentions it multiple times that the event, its causes, and the unknown issues that caused a disproportionate increase in temperature could all be relevant even in the future, because of the high CO2 levels in the current environment. Option A is not the answer.

context mentioned. PETM is being discussed here as an event with a disproportionate increase in temperature, but the reasons are unknown at this point in the passage. Option B is not the answer.
Option C: This is already known that carbon acts as a heat trap. It is also understood that that increase in temperature is way more than expected and that's because carbon alone is not the reason. So, carbon heat trap warming the temperature disproportionately is not a valid conclusion, given there is another hidden parameter. Hence, Option C is not the answer.
Option D: The statement in consideration doesn't really talk about clouds. It only touches upon an x-factor parameter that hasn't been accounted for. So, a conclusion about clouds cannot be arrived at. Hence, Option D is not the answer.
Choice (A)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 551

Option B: This is the reason the author mentions PETM – to draw parallels between the event 56 million years ago and a future possibility of a similar event. It is important because the rising CO2 levels do not on their own explain the massive rise in temperature. It suggests that there was a second, unexpected reason that was triggered during the PETM. Hence, Option B is the answer.

Option C: The outsized influence on earth's climate is not just because of the high CO2 levels. The high CO2 levels beyond a tipping point seem to trigger a second cause (cloud loss as the study found out) which leads to the outsized influence on earth's climate. Option C is not the answer.

Option D: While this is true, it is stating the obvious. The author mentions the PETM not just because they were more extreme than what theoretical models suggest but also to point out that the hidden reason for the unexpected extremes is still relevant. Hence, Option D is close but not the answer.

Choice (B)

8. This can be understood from '***Even after accounting for differences in geography, ocean currents and vegetation during these past episodes***, paleoclimatologists find that something big appears to be missing from their models — ***an X-factor whose wild swings leave no trace in the fossil record.***'

Option A: This is a valid conclusion from the given statements. It can be understood clearly that the theoretical models do not account for all the parameters, and there is another aspect of the environment that was not included in these models. Hence, Option A is the answer.

Option B: A conclusion about the amounts of cloud cannot be made because clouds have not been discussed in the statement mentioned or in the

9. Option A: The author discusses human fertility and mentions that the perspective of biological anthropologists is unique because of the importance they give to evolution and its history. In other words, there is a definite connection between these two. So, it can be inferred that human fertility and human evolution are interrelated. Option A is not the answer.

Option B: The perspective of biological anthropologists is unique (because of two reasons), according to the passage. In other words, it is special, compared to the perspective of other disciplines. The perspective or the opinion itself of biological anthropologists, or what they think, about human fertility need not be more important from what those in other disciplines think. In other words, uniqueness is not the same as importance. Option B is the answer.

Option C: This is quite easy to infer. Given that there is the existence of someone called a 'biological anthropologist' we can infer that biology (species biology as mentioned in the last line of the para) and anthropology are interconnected. Option C is not the answer.

Option D: The first para mentions that the biological anthropologist will have ***a unique perspective because of the central position of evolutionary theory***. So, if this central position makes biological anthropology unique (or the perspective held by someone practising the subject), it means the other disciplines do not give evolution or its history a central position. Option D can be inferred. Option D is not the answer.

Choice (B)

10. Option A: Our evolutionary past is ***both*** the history of change ***and the forces and constraints that shaped it course.*** So, we can see that evolutionary past isn't history of change alone. (Also, 'history of our change' and 'history of

change' aren't the same.) Secondly, this doesn't explain why reproductive biology has been mentioned. Option A is not the answer.

Option B: This option represents the same misinterpretation as the one in Option A, by considering one of the two factors that account for evolutionary past, while also ignoring the importance of reproductive biology. Option B is not the answer.

Option C: This can be understood from: '…*since (because) in its essentials, evolution or natural selection can be broken down into variability in the processes of birth and death*'. So, the author is of the opinion that understanding of reproductive biology is fundamental because evolution can be broken down into (or in other words, evolution includes) variability of processes (of birth and death). Since, birth is involved, we can connect that reproductive biology has a say in evolution. Option C is the answer.

Option D: Evolution has been defined as the variability of the processes of birth and death. It is not hampered by these processes. Rather, it happens because of these processes. 'Broken down' here is not negative. It implies, if we look closely evolution is the sum total of variability. Option D is not the answer. Choice (C)

11. It can be understood from the para that the previous view directly descended from Malthus established a connection between biology and cultural conventions. This was later invalidated. We are looking for an option that doesn't weaken the argument that culture/societal processes/individual choice influence human fertility.

Option A: This option frees human fertility completely from the influence of culture and societal aspects, which were the core parameters in the previous view held by Thomas Malthus and others. So, Option A weakens the 'previous view' and is not the answer.

Option B: Irrespective of whether this statement is true or false, we cannot comment on the view held by Thomas Malthus and others. This option doesn't contradict Malthus' views on the relation between biology/human fertility and cultural conventional/societal pressures. The para mentions that Malthus and others who subscribed to his view believed that biology is manipulated and constrained by culture and '*all the important determinants of variance in human fertility deriving from cultural conventions, societal pressures or individual choice.*' This option doesn't contradict that view. Hence, Option B is the answer.

Option C: This is in direct contrast to the 'previous view' which established that biology and human fertility were a function of cultural conventions/societal pressures and processes and individual choice. Here, it is being said that human fertility is not influenced by cultural factors in any way. Option C weakens the 'previous view'. Option C is not the answer.

Option D: If human physiology's variability is natural, it goes against Thomas Malthus' view that biology is constant and constrained by culture. Here, culture doesn't play a role in biology. Hence, Option D weakens the 'previous view', descended from Thomas Malthus. Option D is not the answer. Choice (B)

12. The benefit of the 'theoretical cohesion' has been mentioned in the last para of the passage: '*Such **an organized theoretical perspective** is necessary to support practical applications of knowledge in this area, as, for instance, in anticipating important policy issues in developing societies, or informing clinical perspectives on fertility and infertility.*'

Option A: The organising of the theoretical perspective or the cohesion happened because of the connection between evolutionary framework and human fertility. In other words, this is the event – the cohesion. The question is about the benefits of this cohesion – how does this organising help. Option A is not the answer.

Option B: This is the benefit as understood from '*Such an organized theoretical perspective is necessary to **support practical applications of knowledge in this area***…'. It informs policy-making as well. Hence, Option B is the answer.

Option C: The last para clearly explains that scientific knowledge can support ***but not validate*** a given policy or determine the course of action. In other words, it can assist in the process of making practical decisions and policies, and help in finetuning them. It cannot really help in determining what is the right course of action and what isn't. Hence, Option C is not the answer.

Option D: A quiet revolution has brought the perspective of biological anthropology to the forefront and that, in turn, has helped in achieving the theoretical cohesion. In other words, the growing importance of biological anthropology is a cause (upstream) rather than a consequence (downstream) of the theoretical cohesion. Option D is not the answer. Choice (B)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 527

13. Option A: The author says that wind and solar power couldn't offer enough to drive the growth. Whatever be the reason for that, this statement is true, as a shortcoming for the standard alternatives. Option A is not the answer.

Option B: This has been directly mentioned by the author that conventional nuclear reactors are massively expensive and dangerous. Option B is not the answer.

Option C: The author mentions that conventional nuclear reactors are an anathema to much of the public. So it can be understood that there is loathing and denunciation/paranoia/dislike for this source of energy. Option C is true and not the answer.

Option D: The author doesn't talk about demand for wind and solar power. It is simply stated that the amount of energy that is needed for the kind of growth required to pull the world of poverty, cannot be supplied by wind and solar power. Whether that is because there isn't enough infrastructure or because we don't have enough scaling opportunity is not understood from the passage. Option D is the answer.
Choice (D)

14. Option A: This is clear since the salts are kept nice and hot by the reaction, the sentence states. So, Option A can be inferred, and is the answer.
Option B: This cannot be understood from the para given that all we know about the salt is that it is green in colour, and undergoes nuclear fission. So, Option B is easy to eliminate. Option B is not the answer.
Option C: While it has been mentioned that the fission takes place inside hot molten liquid, we don't know if this is the only set of conditions where a nuclear fission reaction can take place. Also, the high temperature could be an effect of the reaction as stated in Option A, and need not be required for the reaction to take place. Option C is not the answer.
Option D: While the last sentence mentions the generation of enough heat to power a city or two, nothing has been mentioned about the practically feasibility of the transformation to other forms of energy. We can infer about the magnitude of power generated but not about whether it is practical yet to use that power. Hence, Option D is not the answer. Choice (A)

15. A major chunk of the para passes the message that molten salt reactors cannot be hijacked for misuse, or to make nuclear bombs.
Option A: This contradicts the tone of the last para. In fact, the author tries to explain that it is not possible to hack the system. So, owners making a bomb definitely cannot be inferred from the last para (irrespective of whether it is possible or not). Hence, Option A is not the answer.
Option D: This, shows that molten salt reactors cannot really aid in the process of making bombs, as the contents cannot even be extracted. This can be understood particularly from '*their hot, liquid contents would be very hard for rogue states or terrorists to hijack for making nuclear weapons.*' Option D is the answer. Choice (D)

16. The author's chief claim can be understood from the last line of the passage: that molten salt reactors can make nuclear power the greenest energy source.
Option A: Since it is cleaning up existing nuclear waste, it strengthens the argument that molten salt reactors could make nuclear power a green energy source. Option A is not the answer.
Option B: Since the efficiency is higher, the chances of producing nuclear waste is significantly lower, and adds to the claim that molten salt reactor can be the greenest energy source. Hence, Option B strengthens the author's claim and can be used by the author in support of the main argument. Option B is not the answer.
Option C: This statement signifies the safety mechanism embedded in the process used by molten salt reactors. If the fission products are far more securely trapped, then the chances of catastrophic accidents are significantly lowered. This contributes to the claim that nuclear power can be a green energy source. Option C is not the answer.
Option D: The demonstration of the principle has happened much earlier as the author points out in the first two sentences of the last para. So, the author wouldn't base his argument on the future viability of the technology. This point doesn't strengthen the author's argument, and it is not accurate. Option D is the answer.
Choice (D)

Solutions for questions 17 and 18:

17. Option A: The para doesn't say that the brain ***creates*** consciousness. The para says that the brain ***causes*** consciousness and that is a

Option B: Once again, this goes against the grain of the para. So, irrespective of whether it is true or false, it is not implied by the last para for sure. That's because the last para makes the effort to explain that these molten salt reactors are good, proliferation-resistant and safe. So, such a para cannot imply that they are hard to monitor (which is negative in tone). Hence, Option B is not the answer.

Option C: This directly states that the proliferation resistance is not a benefit compared to normal conventional nuclear reactors. While we don't know what is the danger level of the conventional nuclear reactor, this option only means that we cannot treat proliferation resistance of molten salt reactors as an advantage. That contradicts the tone and message of the para directly where proliferation resistance has been offered as a benefit. Option C is not the answer.

biological phenomenon that we have to accept. Also, the option doesn't reflect the essence of the para – it doesn't explain why we have to acknowledge this phenomenon (to make sense out of what consciousness is). Option A is not the answer.

Option B: This is apt as it indicates the two ideas of the para – that mental phenomena are like any biological phenomena, and we need to identify this to understand the mysterious connection between brain and the consciousness. Option B is the answer.

Option C: This option seems to indicate that the brain and the consciousness are one and the same. That is definitely not what the para means. It also indicates that they are the same metaphysical realm, which is a further distortion. Option C is not the answer.

Option D: This option doesn't say that philosophical and scientific progress has been stalled because the relation between the brain and consciousness hasn't been resolved. The para says progress ***in these areas*** needs us to look at the brain and consciousness (mental phenomena) as biological phenomena. This option, therefore, changes the idea completely. Option D is not the answer. Choice (B)

18. Option A: This option introduces the idea of Oldowan tools and how they are important for human evolution, but there is no clarification about what is meant by 'turning point' in human smartness. That idea that humans began to nurse the idea of a home has not been given. Option A is not the answer.

Option B: This option presents all the important ideas of the para: the Oldowan tools offer insights into the Lower Paleolithic phase of human

Sentence 2 talks about how public-private sector partnerships can solve the drawbacks of private sectors specifically 'to achieve scale and impact.'

Sentence 3 is an independent sentence about technological and operational innovations and how they need to gain momentum across industries.

Sentence 4 refers to 'this partnership' and talks about the individual roles of private and public sectors.

It can be understood that that 4 follows 2 because 2 introduces the partnership between private and public sectors. Also, 2 has to follow 1 because 2 offers the solution to the drawbacks mentioned in 1. 3 is the independent, context-setting sentence that is the biggest of all the ideas here. Hence, the answer is 3124. Ans: (3124)

Solution for question 21:

evolution because these tools enabled that evolution. That phase did involve smarter hominins who were thinking of a sense of home before agriculture had developed. Option B is the answer.

Option C: The Oldowan implements offer insights into a time when hominins were smart and started developing a sense of home, as per the para. However, the para doesn't say that these implements 'establish' the point/timeline specifically at which hominins became smart enough ***to think of settling down***, which is also a bad extrapolation. Option C is not the answer.

Option D: While this option is close, and reasonably correct, it misses out on details like Lower Paleolithic phase, and the hominins developing a sense of home. The option is very broad and, hence, doesn't accurately depict the essence of the para. Option D is not the answer.
Choice (B)

Solutions for questions 19 and 20:

19. Sentence 1 refers to 'the crime' series and 'its popularity'. So, we know 1 must come after a sentence that introduces the series.

Sentence 2 is an independent sentence that explains how popular Breaking Bad was.

Sentence 3 is about Netflix, an introduction to it.

Sentence 4 is about when Netflix purchased Breaking Bad.

It can be understood that 3 must come before 4 because 3 talks about Netflix and what it is, and 4 continues that to what Netflix purchased. Both 1 and 2 are about Breaking Bad, so they come after the 34 block. 2 has to follow 1 because viewers could 'quickly get up to speed' only if Netflix put the first three seasons on its platform. So, 3412 is the order. Ans: (3412)

20. Sentence 1 is an independent sentence about the pros of private sector, and the drawbacks ('limited in scale).

21. Option A: Darwin's sexual selection theory explains the mechanism of complex traits being passed on through generations, with birdsongs being a good example. This option reflects the essence of the para because it mentions all the ideas. Option A is the answer.

Option B: There is a false causation in this option. Birdsongs are not an example of a complex trait ***because*** males use birdsongs to attract mates. It is a complex trait and Darwin thought it is shaped by sexual selection based on how the trait is useful to males. Option B is not the answer.

Option C: While this option mentions both examples in the para, it doesn't use the term 'sexual selection' nor explain the theory as Darwin did. It is too concise to represent the essence of the para. Option C is not the answer.

Option D: This option distorts the idea by saying 'only sexual selection' can explain the non-adaptive characteristics. Such an emphasis has not been placed in the para. It also doesn't explain what such characteristics are or mention Darwin. Option D is not the answer. Choice (A)

Solution for question 22:

22. Sentence 1 talks about scallop dredging and what it is and how it affects marine life, an introductory sentence by nature.

Sentence 2 points to 'this fishery' and talks about its extent in terms of where it is happening.

Sentence 3 is a negative sentence that talks about the warning given by marine scientists about the dangers of bottom trawling.

Sentence 4 talks about another impact 'less publicized' than an earlier mentioned problem. This is the destruction of 'underwater cultural heritage.'

It is obvious that 4 follows 3 because of the comparison. Bottom trawling destroys marine habitats but less publicized impact is the impact on cultural heritage. 1 introduces the topic and 2 gives its extent. 4 and 3 give the consequences. So, the order is 1234. Ans: (1234)

Solutions for questions 23 and 24:

23. Blank 1 precedes a sentence that talks about where the concept comes from and its origins. So, the concept catching on like wildfire in 2010s doesn't make sense here. Similarly, Blank 2 is followed by what Nudge theory suggests, so we get to understand more about it. So, the bold statement won't arrive before we explain what Nudge theory actually is. The sentence preceding Blank 3 provides what nudge theory is and the sentence following blank 3 provides a real-life example of nudge theory. Hence, the bold statement cannot come in blank 3 (primarily because the 'for example' in the following sentence cannot fit with the concept catching on like wildfire). The sentence following blank 4 provides how many institutions started using the theory. This makes blank 4 an apt position for the given sentence. The highlighted sentence talks about the concept becoming highly popular and the sentence following blank 4 provides substantiation for the same. Hence, Option D is the answer. Choice (D)

24. The first blank cannot accommodate the bold statement because there is no context for why we are talking about the future, before we talk about climate change. The second blank is not the right place for the bold statement because it is followed by why it is increasingly harder to be a dictator – explains how political systems are unravelled as mentioned in the first sentence of the para given. The third blank is the right place for the bold statement because it is followed by Keats' advice that 'we should be content with half-knowledge.' Also, there is a shift of tone on either side of the third blank. On the left we discuss how times are changing, and without the transition sentence (the bold sentence), the quotes won't make sense. By the same logic, Blank 4 is not an apt place for the bold statement as the quotes cannot precede this transition statement that changes the tone. Option C is the answer. Choice (C)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	4, 19
Medium	1, 3, 5, 8, 13, 14, 15, 17, 18, 21, 22, 23
Difficult	2, 6, 7, 9, 10, 12, 16, 20, 24
Very Difficult	11

SECTION – II

Solutions for questions 1 to 5:

Given that, on Day 1, the price of Commodity A is 130 and that of Commodity B is 140. The price of A minus the price of B for this day is –10.
On Day 2, the prices of Commodity A and Commodity B will be 160 and 150 respectively. The price of A minus the price of B for this day is 10.
On Day 3, the prices of A and B will be ₹170 and ₹190 respectively. The price of A minus the price of B for this day is –20.
On Day 4, the prices of A and B will be ₹200 and ₹200 respectively. The price of A minus the price of B for this day is 0.
On Day 5, the prices of A and B will be ₹230 and ₹210 respectively. The price of A minus the price of B for this day is 20.
On Day 6, the prices of A and B will be ₹240 and ₹250 respectively. The price of A minus the price of B for this day is –10.
From Day 6 onwards, we can observe that the difference in the prices of the two commodities repeats.

1. During the first five days, the price of Commodity A was greater than that of Commodity B on 2 days (Day 2 and Day 5).
During the next five days, the price of Commodity A was greater than that of Commodity B on 2 days (Day 7 and Day 10).
During the next five days, the price of Commodity A was greater than that of Commodity B on 2 days (Day 12 and Day 15).
Hence, the price of Commodity A was greater than that of Commodity B on 6 days. Ans: (6)

2. The difference between the prices on Day 3 was ₹20. On the eighth, thirteenth, eighteenth day…, the price difference was ₹20. On none of the other days was the difference in price greater than this. Choice (C)

3. The price of each commodity will increase by ₹110 every five days. On Day 4, the prices were ₹200. On Day 9, the prices will be ₹310. Similarly, on Day 19, the price of each stock will be ₹530. On Day 17, the prices of both the commodities will be less than ₹500. Hence, *n* can take a maximum value of 17. Ans: (17)

4. Since we need the differences in the averages, we can add the differences in the prices for 73 days and divide the total difference with 73. For each block of five days, the difference in prices add up to 0.
Hence, from Day 1 to Day 70, the differences will add up to 0.
On Day 71, 72 and 73, the sum of the differences will be - 10 + 10 – 20 = 20
Required average = $\frac{20}{73}$ = 0.274 Choice (B)

5. During the first five days, the price of Commodity B was greater than that of Commodity A on Day 1 and Day 3.
During the next five days, the price of Commodity B was greater than that of Commodity A on two days, Day 6 and Day 8. Hence, the given condition is satisfied on Day 6, Day 11, Day 16…Day

(5k + 1), and on Day 8, Day 13, Day 18... Day (5k + 3).
From the given options, the given condition is satisfied only on Day 93. Choice (C)

Solutions for questions 6 to 10:

From (ii), F could have displayed 53 kg or 61 kg or 64 kg. The actual weight could be 50 kg or 58 kg or 61 kg. From (iii), the actual weight cannot be 50 kg and F could not have displayed 53 kg.

C could have displayed 64 kg or 70 kg and B could have displayed 58 kg or 64 kg. In either case, 64 kg would be displayed by either B or C. Hence, F cannot display 64 kg. Hence, F must have displayed 61 kg and the actual weight must be 58 kg. Since B did not display the actual weight, B must have displayed 64 kg and C must have displayed 70 kg.

From (i), A and D both could not have displayed the actual weight. Hence, A must have displayed 53 kg and D must have displayed 50kg. E must have displayed the actual weight, which is 58 kg.

The following table presents this information:

Machine	Weight
A	53
B	64
C	70
D	50
E (Actual Weight)	58
F	61

6. The actual weight of Himesh is 58 kg. Ans: (58)

7. The difference in weight is 20 kg. Choice (C)

8. Machine A displayed 53 kg. Ans: (53)

9. Machine B would display 26 kg. Choice (D)

10. Two machines, A and D, displayed a lower weight. Choice (B)

Solutions for questions 11 to 15:

From (vi), Karthik collected sheets from F. Hence, F must be in the sixth place.
From (i), C cannot be in the first place. Hence, A must be in the first place and C must be in the second place.
From (iii), (iv) and (v), we know that E, D and B have taken 6, 15 and 8 sheets respectively.
Let the number of sheets taken by A, C and F be 2x, x and y respectively.
Since Karthik gave 60 sheets and took 18 sheets, the students must have taken 60 – 18 = 42 sheets in total.
Hence, $2x + x + y + 6 + 15 + 8 = 42 \Rightarrow 3x + y = 13$
From (iv), x, 2x and y must be greater than 20 or must be less than 10.
From the equation, none of them can be greater than 20.
Hence, all three must be less than 10.
If x = 1, 2x = 2 and y = 10. This is not possible.
If x = 2, 2x = 4 and y = 7. This is possible.
If x = 3, 2x = 6 and y = 4. This is not possible as E took 6 sheets.
If x = 4, 2x = 8. This is not possible as B took 8 sheets.
x cannot be 5 or greater as 2x will fall in the range of 10 to 20.
Hence, only one case is possible: A took 4 sheets, C took 2 sheets and F took 7 sheets.
The following table provides the number of sheets that each student took and the order in which they are sitting:

Student	A	C	E	D	B	F
Number of Sheets	4	2	6	15	8	7

11. C took 2 sheets. Ans: (2)

12. A was sitting to the immediate left of C, the student who took the lowest number of sheets. Choice (A)

13. The difference between the number of sheets that A and D took = 11 Ans: (11)

14. The statement given in option D is true. Choice (D)

15. Four students (E, D, B and F) took more than five sheets. Choice (C)

Solutions for questions 16 to 20:

16. Given that B was the third city that he visited. For B to be the third city, the first two cities can be C-A or C-D or F-D.

Let the positions of the chairs be numbered from 1 to 6 from left to right.
From (v), E must be in the 3rd place and took 6 sheets. From (iii), B cannot be in the 6th place. B also cannot be in the 1st or 2nd place (since there must be a student sitting two places to B's left). Since E is in 3rd place, from (iv), B cannot be in 4th place (as D has to be to B's immediate left).
Hence, B must be in the 5th place and D must be in the 4th place.

If the first two cities are C and A, from B, he has to go to D. From D, he has to go to F. From F, he can go to E or G. If he goes to E, then from E he can go only to one city between G and H. Hence, from F, he has to go to G. From G to E and from E to H. Hence, his route in this case will be C-A-B-D-F-G-E-H.
If the first two cities are C and D, from B he has to go to A. From A, he has to go to C again, which is not possible. Hence, the first two cities cannot be C and D.

If the first two cities are F and D, he has to go from B to A to C to H to E to G. Hence, the route will be F-D-B-A-C-H-E-G.
From the given options, the last city that he visits can be G. Choice (A)

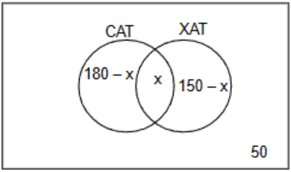
17. If C and D are the last two cities that he visits, then before visiting C and D, he could have visited H, E, G and F OR A and B but not both. Hence, C and D cannot be the last two cities that he visited.
If E and F are the last two cities that he visited, then he could not have visited G (as it is accessible only from E and F). Hence, these cannot be the last two cities that he visits.
If D and F are the last two cities that he visited, he could have visited the cities in the order G-E-H-C-A-B-D-F.
Hence, more than one of the given options cannot be the last two cities that he visited.
Choice (D)

18. If Kiran starts from E, all the cities connected to E can be the last city that he visits. There are 3 cities that are connected to E – F, G and H. Of these, only two cities can be the last city – G and H. F cannot be the last city as it will not be possible to visit G, if F is the last city and E is the first city.
Hence, the answer is 2. Choice (C)

19. If he starts from A, he should not end at B or C. This is possible in two ways – A-B-D-C-H-E-F-G and A-B-D-C-H-E-G-F.
If he starts from B, one way is possible – B-A-C-D-F-G-E-H. If he starts from C, he has to end at H.
For D, one way is possible – D-B-A-C-G-E-F-G.
For E, it is not possible.

As [X] and [Z] are both integers, {Y} is the only part which produces a fractional part $\Rightarrow \{Y\} = 0.2$
$\Rightarrow [X] + [Z] = 11$ ----- (1)
Similarly $[Y] + [X] = 10$ ----- (2), and $\{Z\} = 0.3$
and $[Y] + [Z] = 9$ ------- (3), and $\{X\} = 0.4$
Solving (1), (2) and (3)
$[X] = 6$, $[Z] = 5$ and $[Y] = 4$
$X = [X] + \{X\} = 6 + 0.4 = 6.4$
Similarly, $Y = 4.2$ and $Z = 5.3$
$\therefore [X + Y - Z] = [5.3] = 5$ Ans: (5)

2. The minimum number of students taking both the exams will occur when we consider maximum number of students taking at least one of the exams. We can represent this in a venn diagram as follows:



Now $(180 - x) + (x) + (150 - x) + 50 = 300$
$380 - x = 300$
$\Rightarrow x = 80$
$\therefore N = 80$
The maximum number of students taking both the exams occurs when all those who take XAT also take CAT. We can represent this in a Venn diagram as follows:

For F, one way exists – F-G-E-H-C-D-B-A.
For G, three ways exits – G-E-F-D-B-A-C-H, G-F-E-H-C-D-B-A and G-F-E-H-C-A-B-D.
For H, two ways exist – H-E-G-F-D-C-A-B and H-C-A-B-D-F-E-G.
Hence, the number of possibilities for the first city that he visited is 6 – A, B, D, F, G and H.
Choice (D)

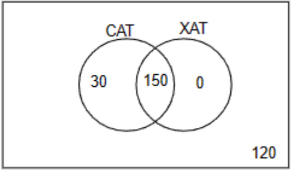
20. From the previous solution, he can visit all the cities in 2 + 1 + 1 + 1 + 3 + 2 = 10 ways.
Choice (A)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	1, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17
Difficult	2, 3, 4, 5, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. Consider the first of the given equations.
[X] + {Y} + [Z] = 11.2



Therefore, the value of M = 150
Thus, M + N = 150 + 80 = 230. Choice (D)

3. $180 = 2^2 \times 3^2 \times 5$
$720 = 2^4 \times 3^2 \times 5$
As L.C.M. of 180 and P is 720, P must be a multiple of 2^4, less than 720 and can have 3 or 5 as other prime factors.
$150 = 2 \times 3 \times 5^2$
$30 = 2 \times 3 \times 5$
As H.C.F. of 150 and P is 30, P must be a multiple, of 30 with only one 5 in it (i.e., P is not a multiple of 5^2) and at least one 3 in it.
From the above, we can say that P can be $2^4 \times 3 \times 5 = 240$ OR $2^4 \times 3^2 \times 5 = 720$.
Hence, the sum of the digits of P is not unique.
Choice (D)

4. The given problem is to find the number of ways of dividing the 8 similar balls into 3 groups with each group having at least one ball.
Thus, the required number of ways is equal to the number of positive integral solutions of the equation.
$x_1 + x_2 + x_3 = 8$ which is equal to ${}^{8-1}C_{3-1} = {}^{7}C_2 = 21$
(Note: the number of positive integral solutions of the equation $x_1 + x_2 + x_3 + \ldots\ldots\ldots x_k = n$ is ${}^{(n-1)}C_{k-1}$)
Ans: (21)

5. Let N = 382 382 ……… upto 300 digits
$\Rightarrow N = 38 \times 10^{298} + 23 \times 10^{296} + 82 \times 10^{294} + 38 \times 10^{292} + \ldots. + 82 \times 10^0$
$\Rightarrow N = 38 \times (10^2)^{149} + 23 \times (10^2)^{148} + 82 \times (10^2)^{147} + 38 \times (10^2)^{146} + \ldots\ldots\ldots. + 82 \times 10^0$

7. Let us initially find the number of ways Sethi can win. If Sethi wins, the match he should be the one who wins the last game and the previous results could be internally arranged to get the number of ways.

Games won by		
Sethi	Wilson	Number of ways
5	0	1
5	1	5!/4! = 5
5	2	6!/4!2! = 15
5	3	7!/4!3! = 35
5	4	8!/4!4! = 70

∴ Sethi can win in 126 ways.
Similarly, Wilson can win in 126 ways. Hence, the match can conclude in 252 ways.

$\Rightarrow \text{Rem}\left[\frac{N}{101}\right] = 38 \times (-1)^{149} + 23 \times (-1)^{148} + \dots + 82$

$\Rightarrow \text{Rem}\left[\frac{N}{101}\right] = 25(-38 + 23 - 82 + 38 - 23 + 82)$

$\Rightarrow \text{Rem}\left[\frac{N}{101}\right] = 0$ Ans: (0)

6. Given the digits P, Q, R, S, T, U, V are 0, 1, 2, 3, 4, 5, 6 (not necessarily in that order) and

$$\begin{array}{r} PQ \\ +\,RS \\ \hline TUV \\ \hline \end{array}$$

The sum of the 2 two-digit numbers is less than 200.
∴ T = 1
Further P ≠ 0, R ≠ 0, Q ≠ 0, S ≠ 0 (∴ if Q or S is 0, V would not be distinct from S or Q and if P or R = 0, then we cannot get a three-digit sum).
∴ U = 0 or V = 0.
If V = 0, {Q, S} = {4, 6}. The greatest value of PQ + RS can only be 56 + 34 and not a 3-digit number.
∴ U = 0
Now {P, R} = {4, 6} because if {P, R} = {4, 5}, PQ + RS can be at the most 46 + 53 = 99 and if {P, R} = {5, 6}, PQ + RS is at least 51 + 62 = 113 while 102 ≤ TUV ≤ 106
Substituting the values in the addition problem we have

$$\begin{array}{l} 4\,- \\ 6\,- \\ \hline 10\,- \\ \hline \end{array}$$

We see that the only possibilities are 42 + 63 = 105 or 43 + 62 = 105. In either case V = 5.
Choice (C)

Alternative Solution:

Consider a string of 5 wins (W's) and 4 Losses (L's). Now these nine games can be arranged in $\frac{(5+4)!}{5!\,4!}$ = 126 ways. These ways include all cases where the match ends after five wins. Hence, there are 126 ways in which Sethi wins and 126 cases in which Wilson wins. Therefore, a total of 252 ways. Ans: (252)

8. $\log_a b = \frac{5}{4}$ and $\log_c d = \frac{3}{2} \Rightarrow b = a^{\frac{5}{4}}$ and $d = c^{\frac{3}{2}}$
For b to be an integer, a must be a perfect fourth power.
Also, for d to be an integer, c must be a perfect square.
∴ Let $a = p^4$ and $c = q^2$ (Given $p^4 - q^2 = 32$)
$(p^2 - q)(p^2 + q) = 32$
Also, $p^2 - q < p^2 + q$
∴ $(p^2 - q, p^2 + q) = (1, 32), (2, 16)$ or $(4, 8)$
$p^2 - q + p^2 + q = 2p^2$
∴ $2p^2 = 33, 18$ or 12
∴ $p^2 = \frac{33}{2}, 9$ or 6
As p^2 must be a perfect square, only possibility is $p^2 = 9$.
∴ $p^4 = 81$
∴ a = 81, c = 49
$b = \left(81^{\frac{1}{4}}\right)^5 = 243$ and $d = \left(49^{\frac{1}{2}}\right)^3 = 343$
d – b = 100 Ans: (100)

9. In ΔABC, 180° – ∠b = ∠a + ∠f + ∠e —— (1)
However, in ΔCBD, 180° – ∠b = ∠g + ∠c —— (2)
From (1) and (2), ∠a + ∠f + ∠e = ∠g + ∠c
⇒ ∠g + ∠c – ∠e = ∠a + ∠f
Hence I = II. Choice (C)

10. Let the SP of one apple be *x*.
SP of 3 apples = 3*x*
This is the same as the profit obtained from selling 5 apples.

∴ Time taken to cover 200k in cycling = 20k units.
∴ Total time spent in evening = 3 × 20k = 60k units.
Total time spent in morning and evening
= 2k + 5k + 20k + 60k = 87k

Profit from selling one apple = $\frac{3x}{5}$

CP of one apple = SP – profit = $x - \frac{3x}{5} = \frac{2x}{5}$

$\therefore$ Profit percentage = $\dfrac{\frac{3x}{5}\times 100}{\frac{2x}{5}} = \frac{3}{2}\times 100 = 150\%$

Choice (C)

11. Let the distance that he totally covered in his morning workout be *d* km. Distance he covered in each of swimming, running and cycling will be $\frac{d}{3}$ km. Let his speed of cycling be 10*x* kmph. His speeds of running and swimming are 4*x* kmph are *x* kmph respectively.

His total workout time in the morning

$$= \frac{\left(\frac{d}{3}\right)}{10x} + \frac{\left(\frac{d}{3}\right)}{4x} + \frac{\left(\frac{d}{3}\right)}{x} = \frac{9d}{20x} \text{ hours}$$

His total distance covered in the evening workout = 5*d* km. Let his time spent for each activity then be *t* hours.

$(10x + 4x + x)\, t = 5d$

$t = \frac{d}{3x}$

$\therefore$ Total time spent = $\frac{9d}{20x} + 3t = \frac{9d}{20x} + 3\left(\frac{d}{3x}\right) = \frac{29d}{20x}$

= 348 min (given).

Now, the total time spent on cycling

$= \frac{d}{30x} + \frac{d}{3x} = \frac{11d}{30x}$

$= 348 \times \frac{20}{29} \times \frac{11}{30}$ = 88 minutes

Alternative Solution:

Let the distance be the LCM of the speeds, i.e., 20k.

Since the speeds are in the ratio 10 : 4 : 1, the time taken is in the ratio 2k : 5k : 20k in the morning for cycling, running and swimming respectively.

Total distance covered in the morning = 3 × 20k = 60k.

$\therefore$ Total distance covered in the evening = 5 × 60k = 300 k.

Since the time spent in evening on each of the three activities is the same, distance covered in cycling, running and swimming is in the ratio of speeds, i.e., 10 : 4 : 1.

$\therefore$ Distance covered in evening in cycling

$= \frac{10}{15} \times 300$ k

= 200k.

Time taken to cover 20k in cycling is 2k units.

Given 87k = 348 $\Rightarrow$ k = 4

$\therefore$ Total time spent in cycling = 2k + 20k = 22k = 88 minutes

Ans: (88)

12. As $q - p = r - q$, $2q = p + r$.

$\therefore p + q + r = 3q$

When the product of two or more positive numbers is constant, their sum (3*q*, in this case) will be minimum (hence *q* will be minimum) when all the numbers are equal.

As *pqr* is a constant, $p + q + r$ will be minimum when $p = q = r = 6^{\frac{1}{3}}$.

$\therefore$ When $q = 6^{\frac{1}{3}}$, $3q$ and hence *q* will be minimum.

$\therefore$ The minimum value of *q* is $6^{\frac{1}{3}}$.

Choice (C)

13. Substituting $p = 3q$ and $m = \frac{5n}{3}$ in each of A and B gives

$A = (3q)(q)\left(\frac{5n}{3}\right)^2 + 3q^2n^2 = \frac{34}{3}\, q^2n^2$ and

$B = (3q)^2\left(\frac{5}{3}n\right)n + 3\left(\frac{5}{3}n\right)(3q)nq = 30q^2n^2$

$$\Rightarrow \frac{A+B}{A-B} = \frac{\frac{34}{3}q^2n^2 + 30q^2n^2}{\frac{34}{3}q^2n^2 - 30q^2n^2} = \frac{n^2q^2(34+90)}{n^2q^2(34-90)}$$

$= \frac{124}{-56} = \frac{-31}{14}$

$\therefore \left|\frac{A+B}{A-B}\right| = \frac{31}{14}$

Choice (A)

14. Since k and 3k are roots of $cx^2 + ax + b = 0$, we can consider a, b, c as unknowns and write two equations

$(k^2)c + (k)a + (1)b = 0 \quad \rightarrow \quad (1)$

and $(9k^2)c + (3k)a + (1)b = 0 \quad \rightarrow \quad (2)$

For any two equations $a_1x + b_1y + c_1z = 0$ and $a_2x + b_2y + c_2z = 0$,

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_1c_2 - a_2c_1} = \frac{z}{a_1b_2 - a_2b_1}$$

Now from (1) and (2), a, b, c can be related as

$$\frac{c}{3k - k} = \frac{a}{-(9k^2 - k^2)} = \frac{b}{9k^3 - 3k^3}$$

= m (say) $\rightarrow$(3)

Now substituting for values of a, b, c from (3) into the equation given in the question,

i.e., 3(b – a) = 5c,

We get $3(3mk^3 + 4mk^2) = 5(2mk)$

$\Rightarrow 9k^2 + 12k - 5 = 0$

$\Rightarrow k = \frac{+2 \pm \sqrt{144 + 180}}{18} = \frac{-12 + 18}{18}\ (\because k > 0)$

$\Rightarrow k = \frac{1}{3}$ and $(k)^{\frac{1}{k}} = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$.

Alternative solution:

$3(b - a) = 5c$ can be re-written as

$\frac{5}{3} = \frac{b}{c} - \frac{a}{c} \quad \rightarrow \ (1)$

In $cx^2 + ax + b = 0$, sum of roots $= 3k + k$

$= \frac{-a}{c} = 4k$

Product of roots $= 3k^2 = \frac{b}{c}$. Now, it can be observed that $\left(\frac{b}{c}\right)$ and $\left(-\frac{a}{c}\right)$ can be directly plugged into eq (1) and solved for k.

$\therefore 3k^2 + 4k = \frac{5}{3}$

$\Rightarrow 9k^2 + 12k - 5 = 0$

$\Rightarrow k = \frac{-12 + \sqrt{144 + 180}}{18} = \frac{-12 + 18}{18} = \frac{1}{3}$

(since $k > 0$)

$\therefore k^{1/k} = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$ Choice (B)

15. It is required to find the remainder when 32^{1000} is divided by 97.
Let n = 32 and p = 97 (97 is a prime number) further n is not divisible by p.
Hence $32^{97-1} - 1$ is divisible by 97.
$\Rightarrow 32^{96}$ will leave a remainder of 1 when divided by 97.
Now, $32^{1000} = 32^{96 \times 10} \times 32^{40}$
$\Rightarrow$ the required remainder = remainder when $(1)^{10} \times 32^{40}$ is divided by 97
$32^{40} = (2^5)^{40} = 2^{200} = 2^{96 \times 2} \times 2^8$ and since 2 is also not divisible by 97, 2^{96} when divided by 97 will leave a remainder of 1.
$\therefore$ required remainder = remainder when $(1)^2 \times 2^8$ is divided by 97, i.e., $256 \div 97$ leaves a remainder of 62. Choice (D)

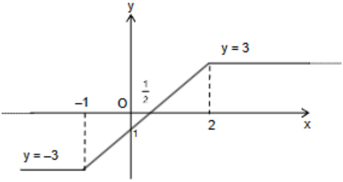
16. Let us suppose that each kg of sugar costs ₹1
So, for ₹100 he should get 100 kg but he gets 120 kg.
Let us assume he is selling all the 120 kg, provided the customer asks for 120 kg + 30% (120 kg) = 156 kg. Hence, the customer pays ₹156. In addition to this he charges 10% of what he has invested on the actual sugar delivered, i.e., 100 i.e., ₹10 as transportation charges. So, he sells the entire 120 kg for ₹166.
$\therefore$ Gain% = 66% Choice (A)

17. As the logs are in AP, the numbers must be in G.P i.e., $3(3^x + 13/3) = 2^{2x}$
$\Rightarrow 3^{x+1} + 13 = 2^{2x}$
So, we can say that $1 < x < 5$.

Trying x = 2, 3, 4,
$x = 2 \Rightarrow 3^{x+1} + 13 = 40$ and $2^{2x} = 16$
$x = 3 \Rightarrow 3^{x+1} + 13 = 94$ and $2^{2x} = 64$
$x = 4 \Rightarrow 3^{x+1} + 13 = 256$ and $2^{2x} = 256$
$\therefore x = 4$ Ans: (4)

18. The price of the chemical X is ₹150 per litre.
The price of the chemical Y is ₹300 per litre.
The weight per litre of X and Y are 12 kg / litre and 15 kg / litre.
For ₹300 the chemical X can have 24 kg or the chemical Y can have 15 kg.
$\therefore$ To have the maximum weight in the solution, the chemical Y must be as less as possible.
But the least quantity we can have is 20% of the total weight of the solution.
Let x kg and y kg be the weight of the solutions X and Y respectively.
$\therefore 20\% \, (x + y) = y \Rightarrow \frac{x}{5} + \frac{y}{5} = y \Rightarrow x = 4y$
$\therefore$ total price $= \frac{x}{12}(150) + \frac{y}{15}(300)$
$= \frac{4y}{12}(150) + \frac{y}{15}(300) = 50y + 20y =$ ₹$70y$
But the total price = ₹2100
$\therefore y = \frac{2100}{70} = 30$
$\therefore x = 4y = 4(30) = 120$
$\therefore$ total weight = 30 + 120 = 150 kg. Choice (C)

19. The plot of $y = |x + 1| - |x - 2|$ is



So, the range of $y = -3 \le y \le 3$.

Alternative Solution:

Since $|x - a|$ is simply the distance between 'x' and 'a' on the number line, $y = |x + 1| - |x - 2|$ is (the distance of x from '–1') minus (the distances of x from '2').
Clearly for all $x \le -1$, this difference is –3 and for all $x \ge 2$, it is 3.
For $-1 < x < 2$, the difference is less than 3.
Hence, $-3 \le y \le 3$. Choice (B)

By observation
$x = 1 \Rightarrow 3^{x+1} + 13 = 22$, which is greater than $2^{2x} = 4$
$x = 5 \Rightarrow 3^{x+1} + 13 = 742$, which is lesser than $2^{2x} = 1024$

20. Treat the given expression as the sum of two independent terms and minimize each term.

Consider $\frac{x}{y} + \frac{4y}{x}$

Here $\frac{x}{y}\left(\frac{4y}{x}\right) = 4$

So, $\frac{x}{y} + \frac{4y}{x}$ will be minimum when $\frac{x}{y} = \frac{4y}{x}$

$\Rightarrow \frac{x}{y} = 2$ ------ (1)

Now consider $\frac{z}{12x} + \frac{x}{3z}$

Here $\frac{z}{12x}\left(\frac{x}{3z}\right) = \frac{1}{36}$

So $\left(\frac{z}{12x} + \frac{x}{3z}\right)$ will be minimum when $\frac{z}{12x} = \frac{x}{3z}$

$\Rightarrow \frac{x}{z} = \frac{1}{2}$ ------- (2)

(2) ÷ (1) gives $\frac{y}{z} = \frac{1}{4} = 0.25$ Ans: (0.25)

21. The amount repaid by Amar to RIB

$= 15000 \times \left(1 + \frac{20}{100}\right)^2 = 15000 \times \frac{36}{25}$

Amount repaid by Amar to SIB

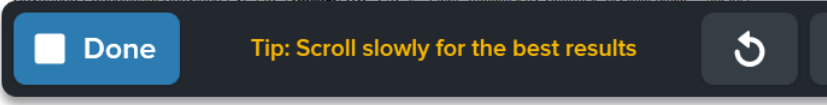
$= 15000 \times \left(1 + \frac{10}{100}\right)^{2 \times 2}$

$= 15000 \times \left(\frac{11}{10}\right)^4$

Required ratio $= \frac{36}{25} \times \left(\frac{10}{11}\right)^4 = \frac{36 \times 400}{11^4} = \frac{14400}{14641}$

Choice (A)

22. Any power of any number ending with 6 (with index greater than 1) always ends with a 6 and also the tens digit of that power is always odd.
Hence 160^{4320} has 4320 zeros, a six before that and an odd number before that. Hence, we encounter the first odd number after 4321 digits, if we start from the right. Ans: (4321)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	10, 13, 16, 21
Medium	1, 2, 3, 4, 6, 9, 12, 17, 19, 22
Difficult	5, 7, 8, 11, 14, 15, 18, 20
Very Difficult	–